ASSIGNMENT: 3

Task (1): List 5 common BIOS or POST error messages you might see when starting a PC and write what hardware or configuration issue each one usually points to.

Ans.

BIOS or POST error messages:

1. **CMOS Checksum Error:**
   - Weak/dead CMOS battery or corrupted BIOS/CMOS settings.
2. **Keyboard Error/Keyboard Not Detected:**
   - Keyboard is disconnected, faulty, or the USB/PS/2 port has a problem.
3. **No Boot Device Found:**
   - HDD/SSD is not detected, damaged, or the boot order is incorrect.

4. **Memory Error/RAM Error:**
    - ➢ RAM is loose, incompatible, or faulty.

5. **CPU Fan Error:**
    - ➢ CPU fan is disconnected, not spinning, or the BIOS detects an incorrect fan speed.

<u>Task (2):</u> Given this scenario: On boot, a PC beeps 3 times in a repeating pattern and does not display anything on the screen. Research and write what this POST beep code typically means for an AMI BIOS, and suggest two troubleshooting steps.

<u>Ans.</u>

AMI BIOS POST Beep Code:

- ➢ For a traditional AMI BIOS, 3 short beeps typically indicate a base 64 KB RAM failure / main-memory read-write test error. This means the BIOS is detecting a problem with the system memory during POST.

❖ Two troubleshooting steps:

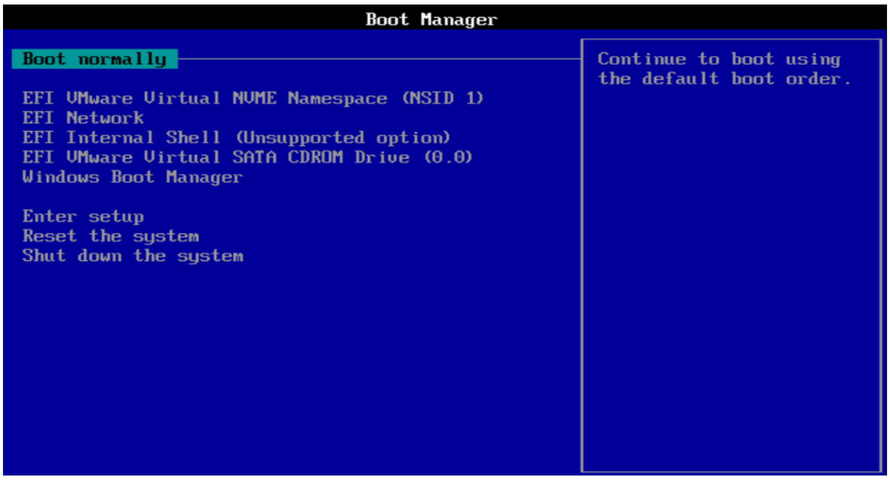
1) Reseat the RAM

- Turn off and unplug the PC.
- Remove the RAM module(s), clean the contacts gently if necessary, and firmly reinstall them.
- Try booting again.

2) Test RAM modules individually

- If there are two or more RAM sticks, remove all but one and try booting.
- Repeat with each stick and, if needed, try different memory slots.
- This can help identify a faulty RAM module or motherboard memory slot.

Task (3):



BIOS Version

BIOS Version/Date	VMware, Inc. VMW201.00V.24006586.BA64.2406042154, 04-06-2024

BIOS Motherboard Manufacturer

BaseBoard Manufacturer	VMware, Inc.

Task (4):

“CMOS checksum error – Defaults loaded”

Meaning:

“CMOS checksum error – Defaults loaded” means the BIOS detected that the settings stored in the CMOS memory are incorrect, corrupted, or have been lost.

How to resolve it in a real system:

1. Restart the computer and enter the BIOS/UEFI setup using a key such as Delete, F2, or the key shown during startup.

2. Check the date, time, boot order, and other important BIOS settings.
3. Select Load Optimized Defaults/Load Setup Defaults, then save and restart.
4. If the error returns repeatedly, shut down the PC, disconnect power, and replace the CMOS battery (CR2032) on the motherboard.
5. Start the PC, enter BIOS again, set the correct date/time and boot settings, and save the changes.
6. If the problem continues even after replacing the battery, check for BIOS corruption or a possible motherboard fault.